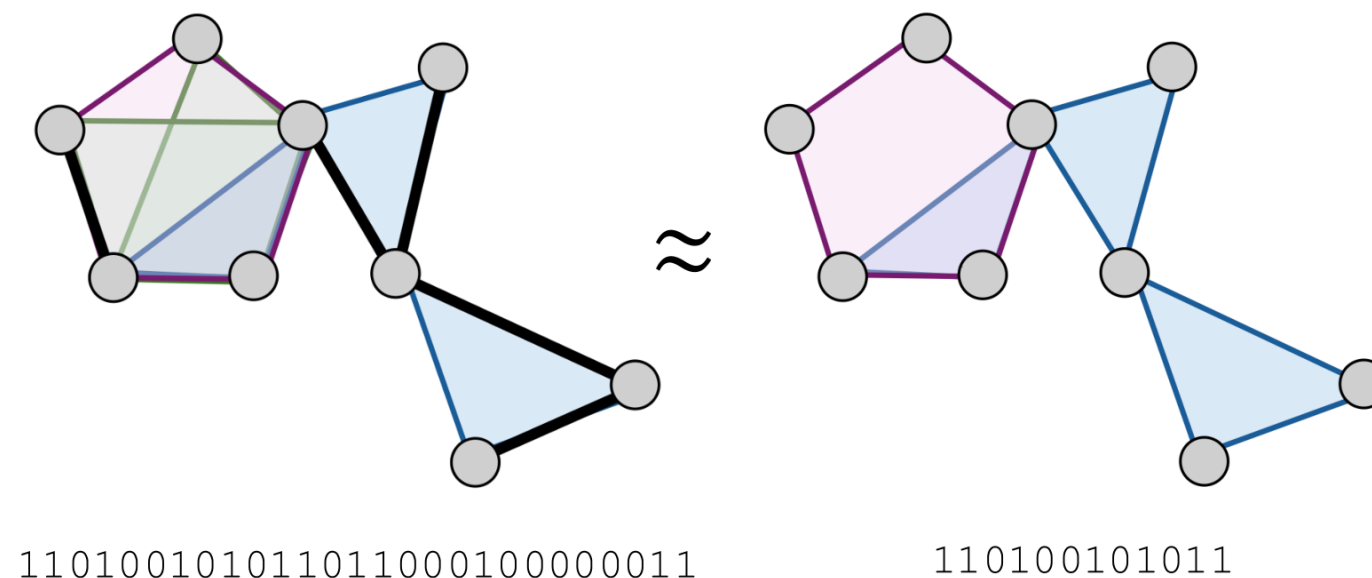
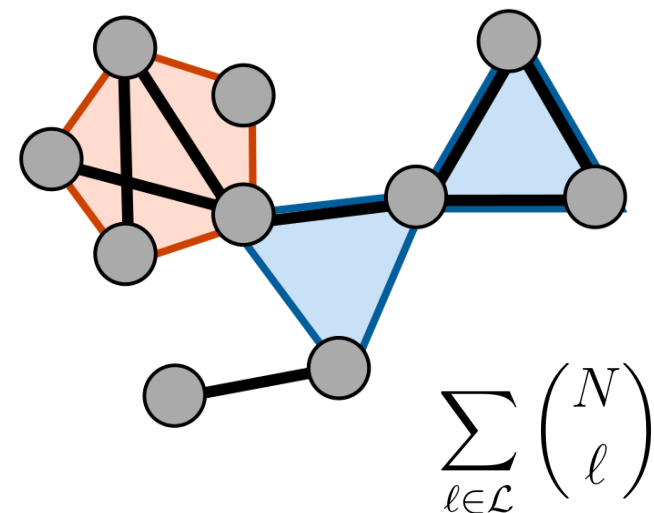
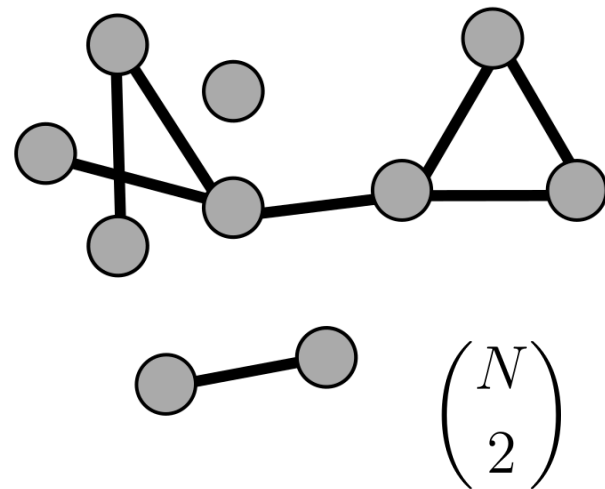


Structural reducibility of hypergraphs

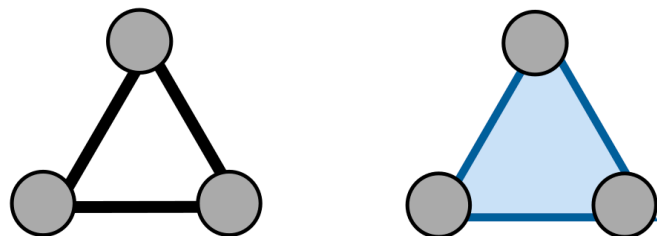


Representational challenge

Computational cost; number of possible hyperedges grows explosively.



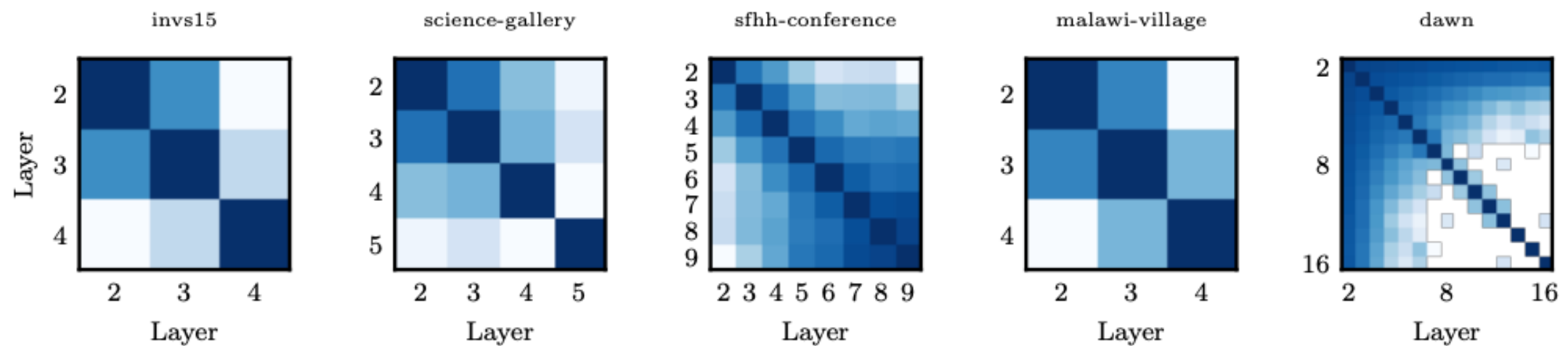
Interpretation; which nested interactions are by-products of higher-order ones?



(three 2-body or one 3-body?)

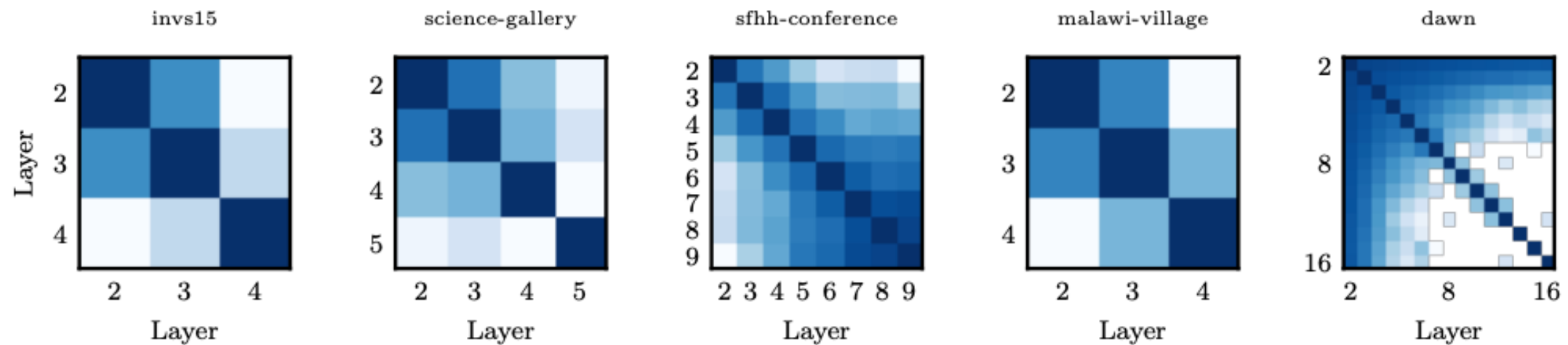
An empirical motivation

Observed nestedness in real-world hypergraphs.



An empirical motivation (and a shameless plug)

Observed nestedness in real-world hypergraphs.



Join us Thursday morning:



PS 2.3 Higher-order & Multilayer Networks 2

📅 Thursday, June 4 | ⌚ 11:00 AM – 12:30 PM | 📍 Central Square

12:15 PM Information theory for hypergraph similarity
Helcio Felipe

MDL principle and network entropies

Best model is that with the shortest description.

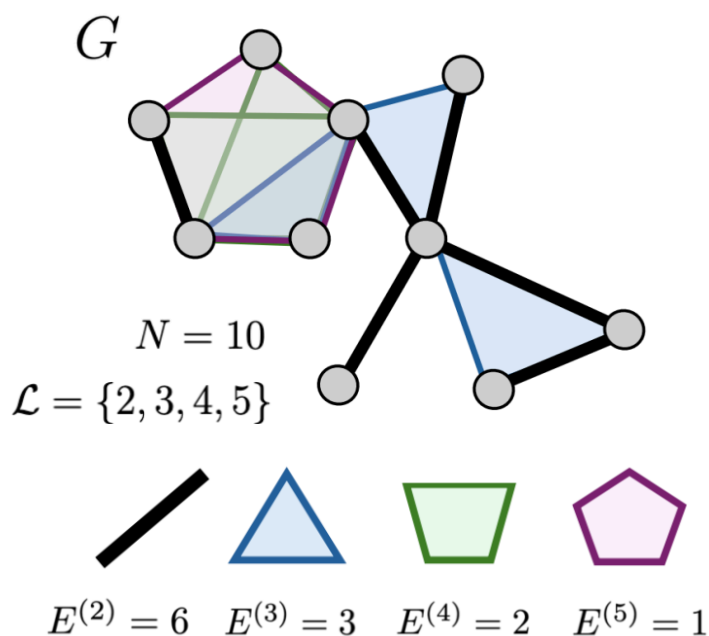
$$DL = H(\text{Model}) + H(\text{Data} \mid \text{Model})$$

MDL principle and network entropies

Best model is that with the shortest description.

$$DL = H(\text{Model}) + H(\text{Data} \mid \text{Model})$$

Information is computed via entropy code-lengths conditioned on previously transmitted constraints (Kraft inequality).



$$\begin{aligned} H(G) &= \log (\# \text{ possible } G \mid N, \mathcal{L}, \{E^{(\ell)}\}) \\ &= \sum_{\ell \in \mathcal{L}} \log \binom{N}{E^{(\ell)}} \end{aligned}$$

Computing conditional entropies

Amount of information required to transmit G_1 given that G_2 and its overlap with G_1 are known.

“Overlap” is where we impose our inductive biases.
For nestedness analysis, overlap is intersection of projected layers.

$$H(G^{(\ell)} \mid G^{(k)}) = \log \binom{E^{(k \rightarrow \ell)}}{E^{(k \cap \ell)}} \binom{\binom{N}{\ell} - E^{(k \rightarrow \ell)}}{E^{(\ell)} - E^{(k \cap \ell)}}$$

($G^{(k \rightarrow \ell)}$ is projection of all tuples of size k in G onto their nested tuples of size ℓ)

Reducibility encoding

A subset R of “representative” layers of G is a good summary of G if it is easily transmitted from R .

$H_G(\mathcal{R})$ = information to describe G using $\mathcal{R} \subseteq \mathcal{L}$

$$= \sum_{r \in \mathcal{R}} H(G^{(r)}) + \sum_{\ell \in \mathcal{L} \setminus \mathcal{R}} H(G^{(\ell)} | G^{(r(\ell) \rightarrow \ell)})$$

(info to transmit rep. layer)

(info to transmit non-rep. layer from its rep. layer)

Reducibility encoding

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(info to transmit rep. layer)

(info to transmit non-rep. layer from its rep. layer)

$$r(\ell) = \operatorname{argmin}_{r \in \mathcal{R}, r > \ell} \{H(G^{(\ell)} | G^{(r)})\}$$

(rep. layer is the “closest” to ℓ)

$$\mathcal{R}^* = \operatorname{argmin}_{\mathcal{R} \subseteq \mathcal{L}} \{H_G(\mathcal{R})\}$$

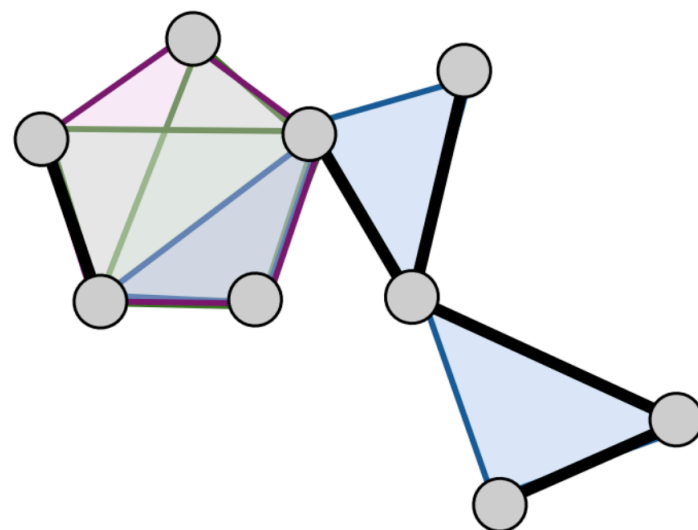
(optimal representative according to MDL)

Structural reducibility

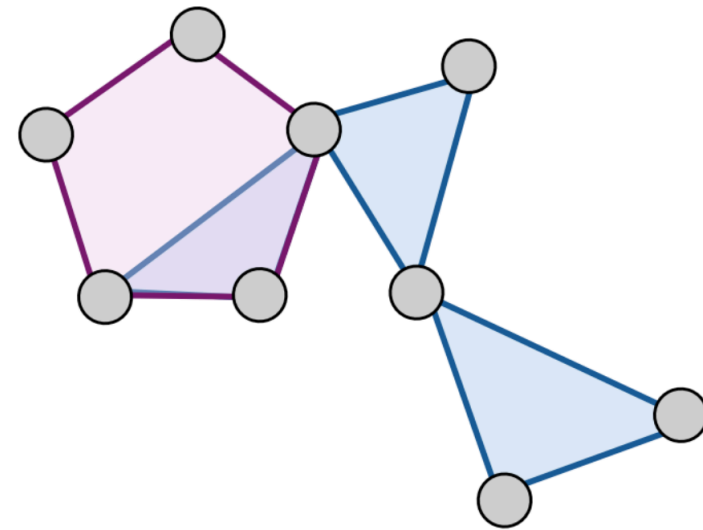
Fraction of redundant information (“reducibility”).

$$\eta = \frac{H_0 - H_G(\mathcal{R}^*)}{H_0 - H_{\ell_{\max}}} \in [0,1]$$

Original hypergraph



Reduced hypergraph







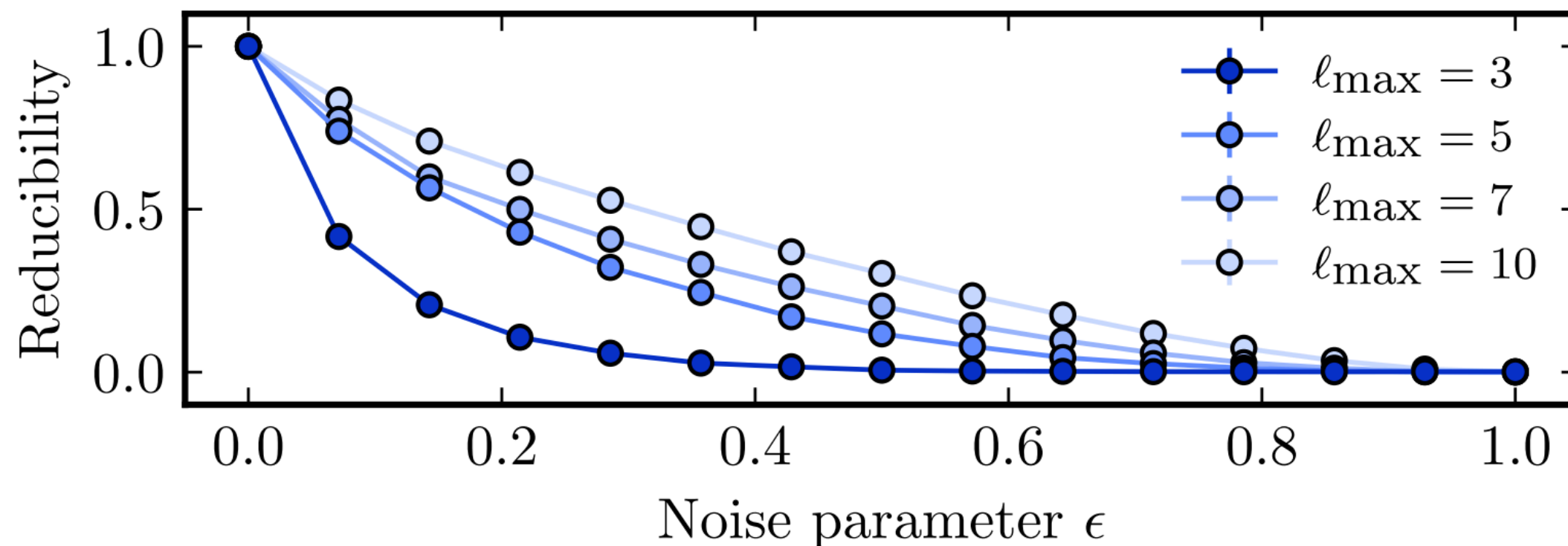

$$E^{(2)} = 5 \quad E^{(3)} = 3 \quad E^{(4)} = 2 \quad E^{(5)} = 1$$

$$\eta = 0.34$$

$$\mathcal{R}^* = \{3, 5\}$$

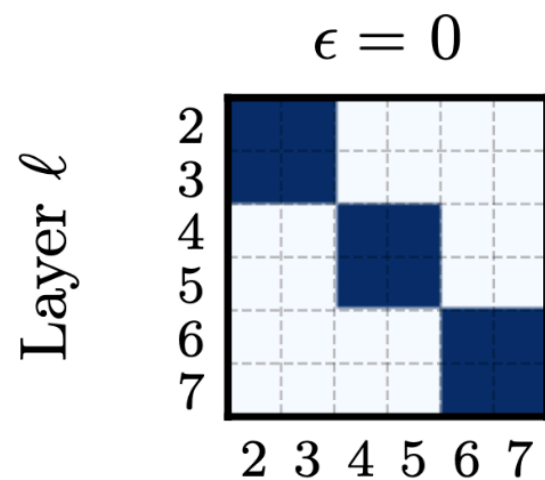
Reducibility of nested hypergraphs

Reducibility decreases with edge randomization, with steeper declines for smaller simplicial complexes.



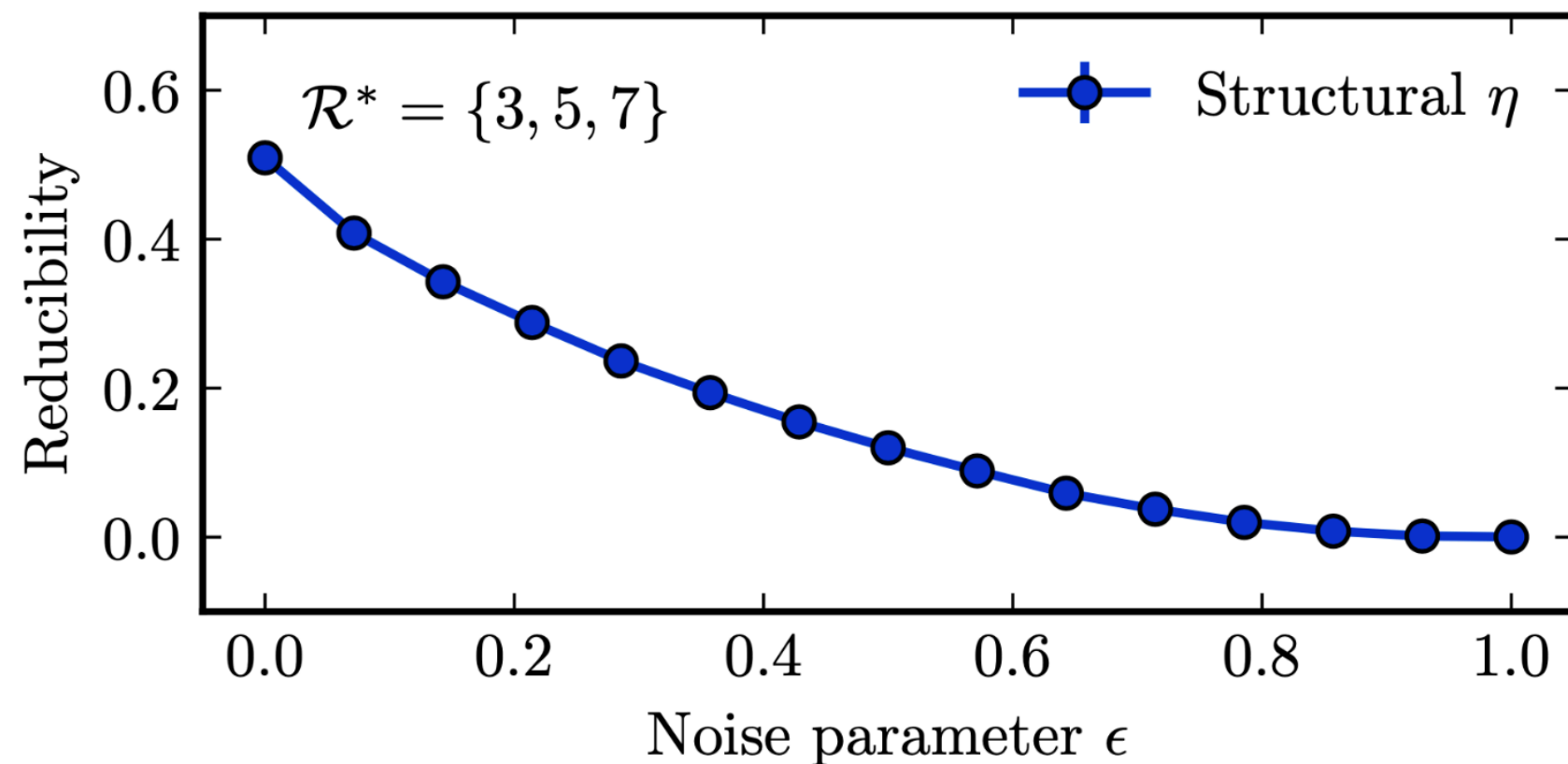
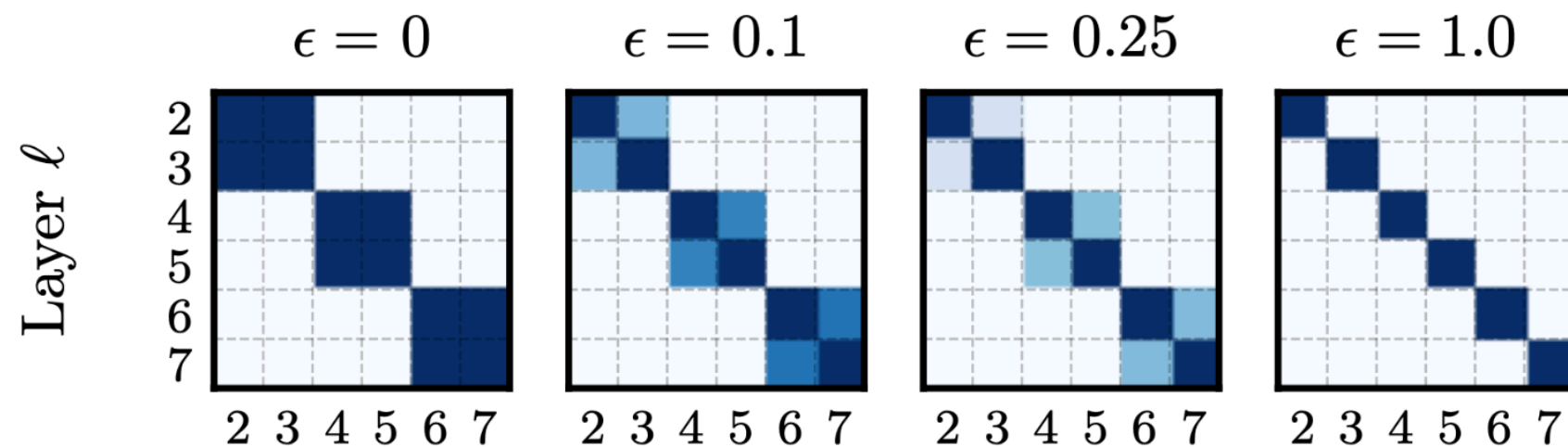
Block-nested hypergraphs

Recovery of planted optimal representative layer.



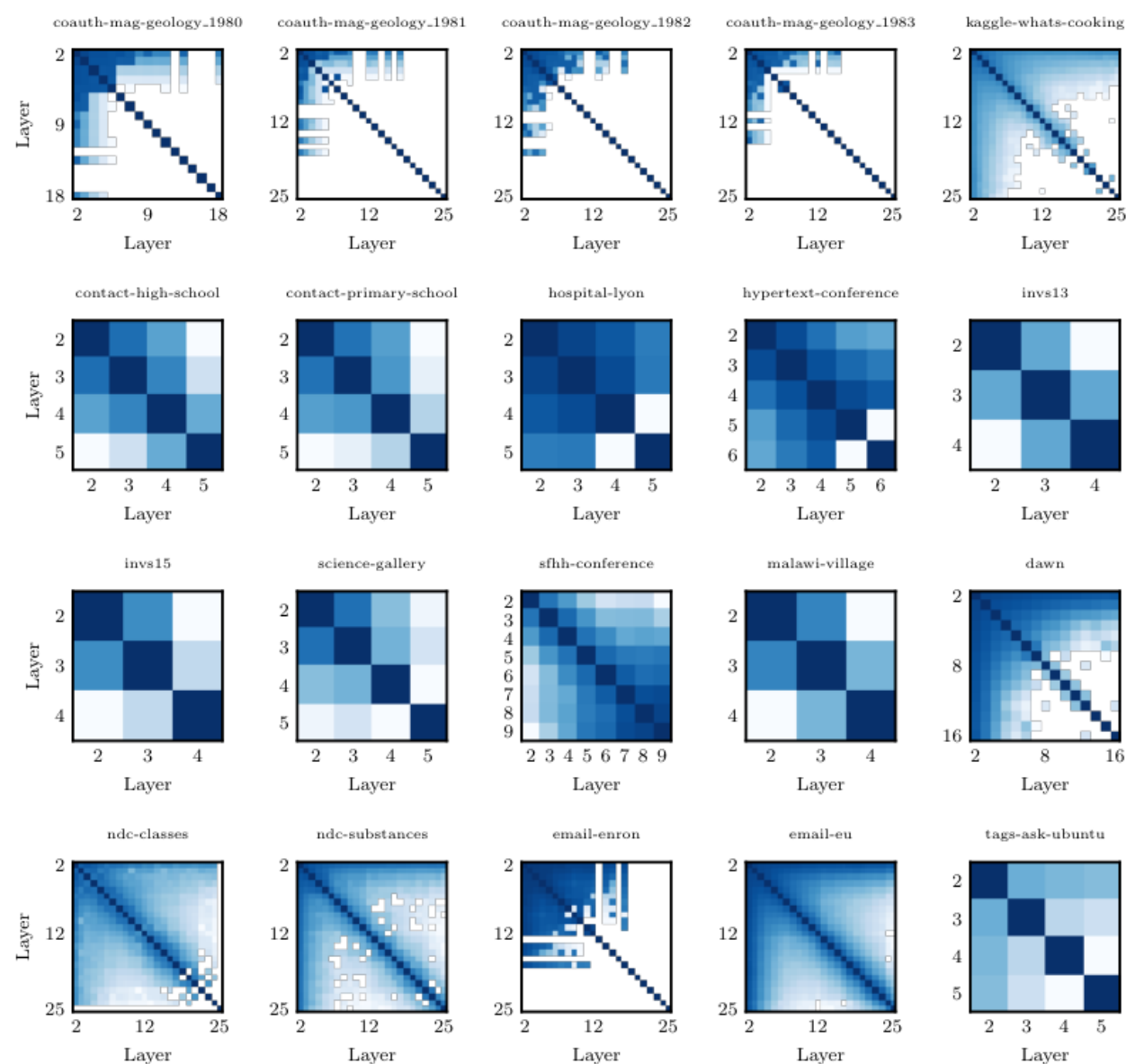
Block-nested hypergraphs

Recovery of planted optimal representative layer.



Reducibility of real-world hypergraphs

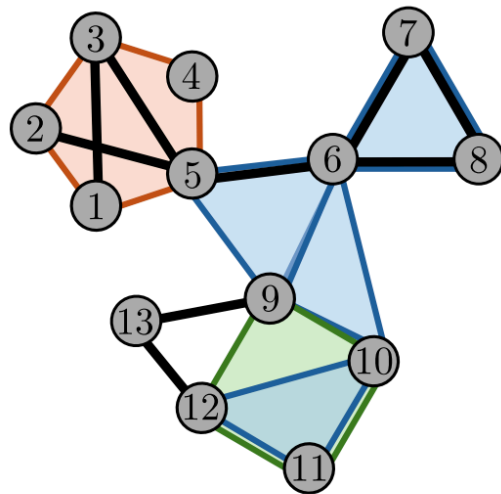
Nested architecture (left) and reducibility (right).



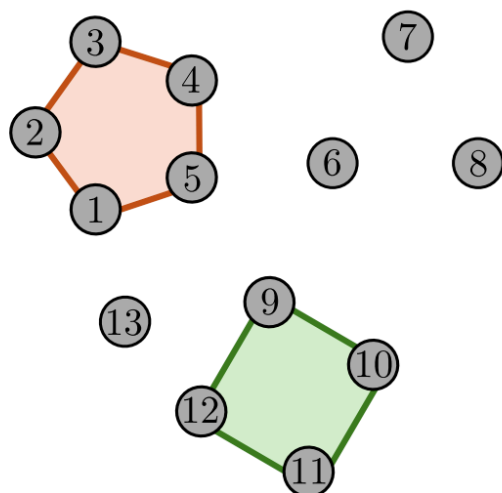
Dataset	N	E	$\ell_{\max}$	$\mathcal{R}^*$	η
coauth-mag-geology_1980	1674	903	18	{3, 5, 9, 18}	0.03
coauth-mag-geology_1981	1075	547	29	{4, 5, 8, 29}	0.01
coauth-mag-geology_1982	1878	987	26	{4, 6, 26}	0.02
coauth-mag-geology_1983	1734	883	36	{4, 36}	0.03 [†]
kaggle-whats-cooking	6714	39224	65	{6,8,9,11,65}	0.04 [†]
contact-high-school	327	7818	5	{3, 5}	0.13
contact-primary-school	242	12704	5	{4, 5}	0.09
hospital-lyon	75	1824	5	{4, 5}	0.11
hypertext-conference	113	2434	6	{3, 5, 6}	0.06
invs13	92	787	4	{3, 4}	0.05
invs15	217	4909	4	{3, 4}	0.10
science-gallery	410	3350	5	{3, 5}	0.16
sfhh-conference	403	10541	9	{4,9}	0.10
malawi-village	84	431	4	{3, 4}	0.19
dawn	2290	138742	16	{6,7,13,16}	0.15
ndc-classes	628	796	39	{4, 9, 10, 13, 14,23,27,39}	0.40 [†]
ndc-substances	3414	6471	187	{5, 40, 187}	0.31 [†]
email-enron	143	1459	37	{4,6,11,12,37}	0.17 [†]
email-eu	986	24520	40	{5,7,8,9,10,11, 12,13,27,39,40}	0.19 [†]
tags-ask-ubuntu	3021	145053	5	{5}	0.17
tags-math-sx	1627	169259	5	{5}	0.26

Hypergraph backboning

Filtering method to capture local heterogeneities.



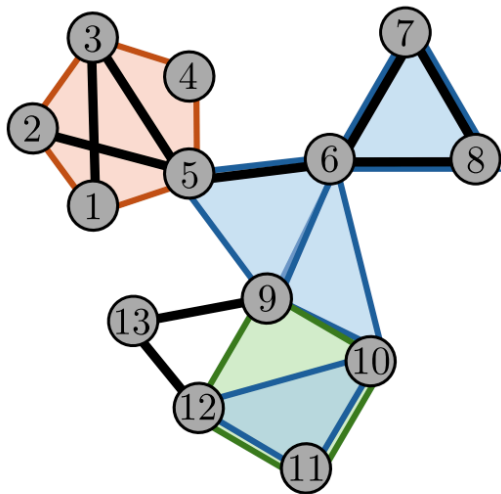
Original hypergraph G



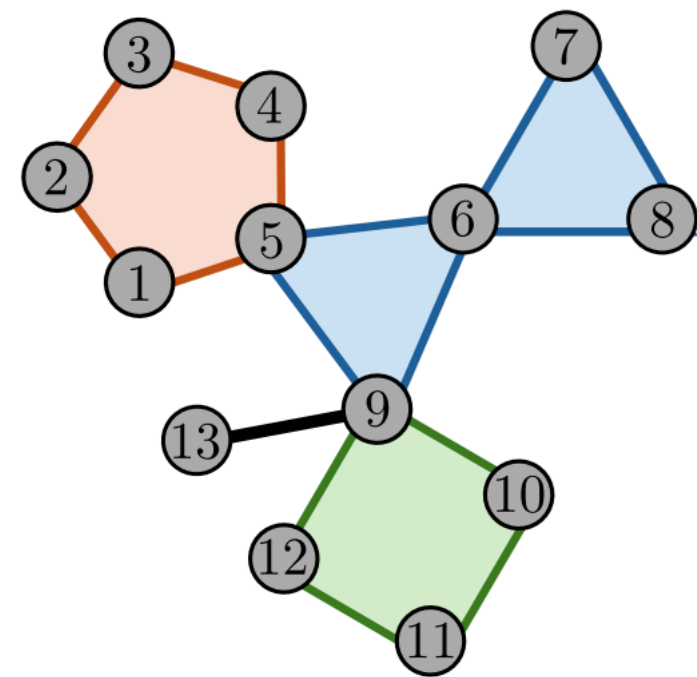
Reduced R^*

Hypergraph backboning

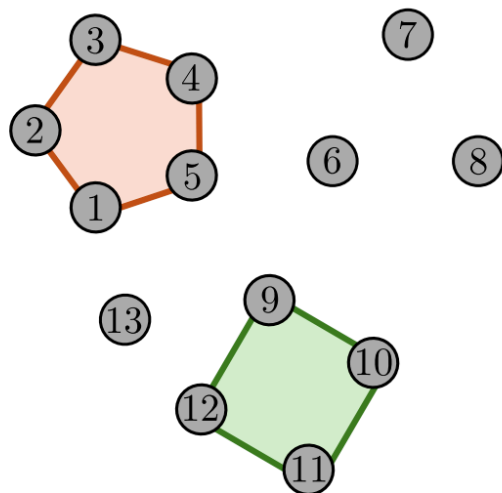
Filtering method to capture local heterogeneities.



Original hypergraph G



Backbone B^*



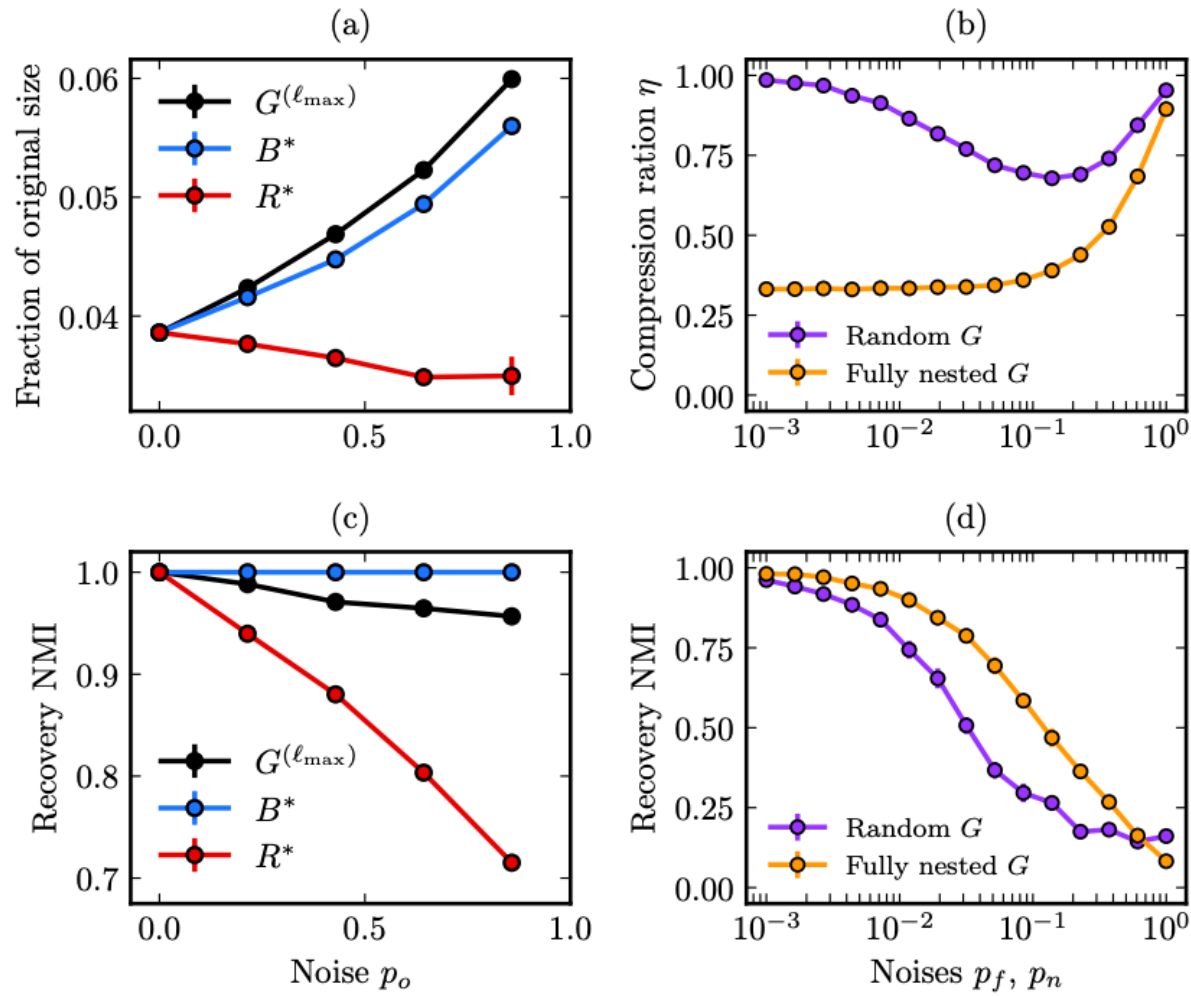
Reduced R^*

Optimal backbone: $B^* = \arg \min_{B \subseteq G} \{\mathcal{L}(G, B)\}$

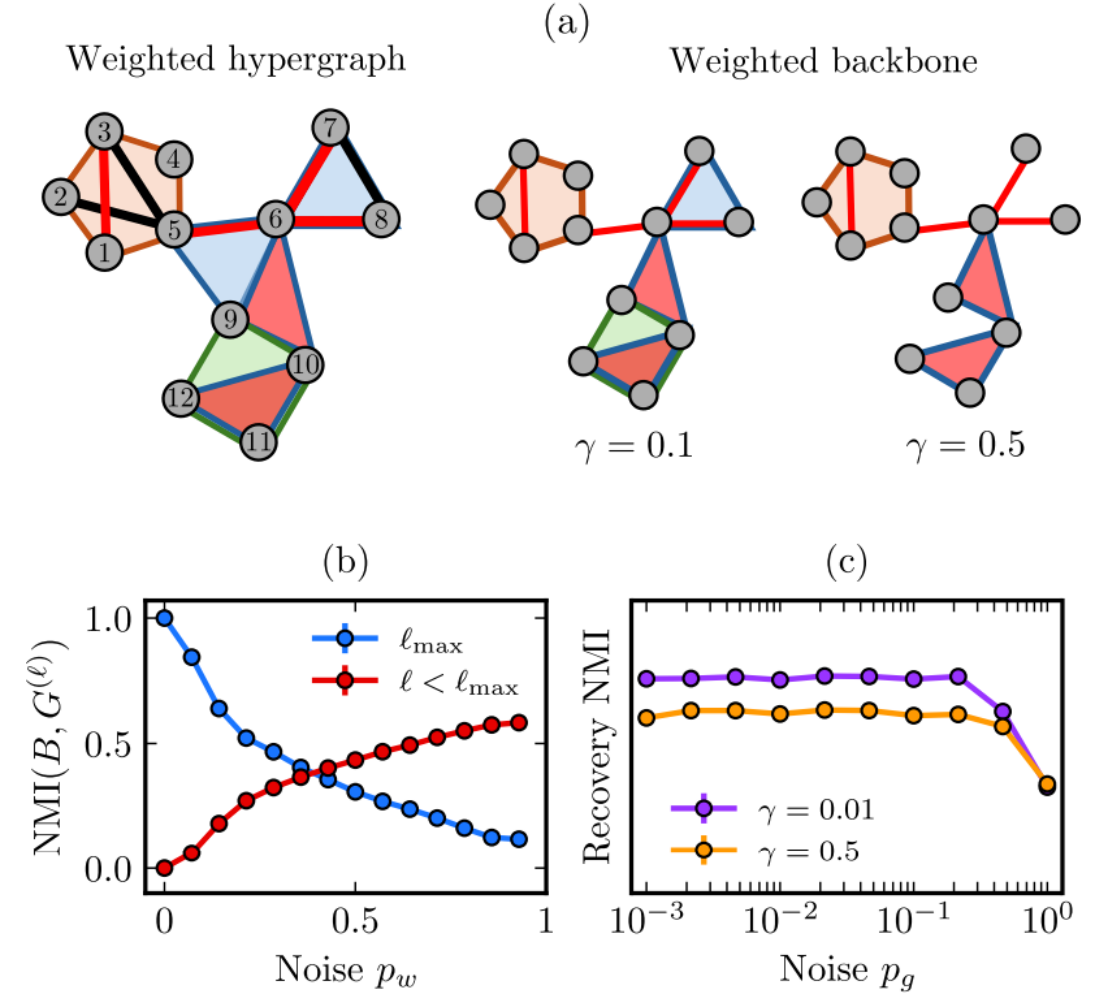
Description length objective: $\mathcal{L}(G, B) = \sum_{p \in B} \left[H(p) + \sum_{c \in \partial_p} H(c | p) \right]$

Compression ratio: $\eta = \frac{\mathcal{L}(G, B^*)}{\mathcal{L}(G, G)}$

Synthetic backbone experiments



(backbone B^* is better at recovering simplicial complexes with varying-level of heterogeneity)



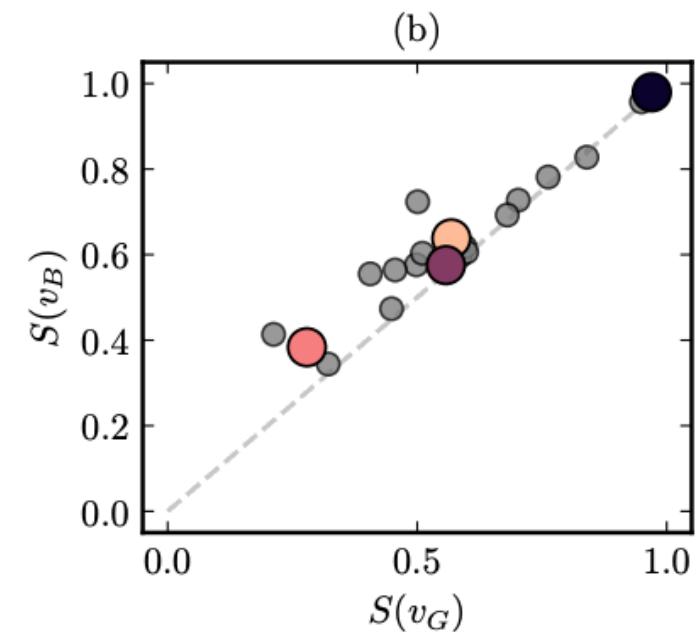
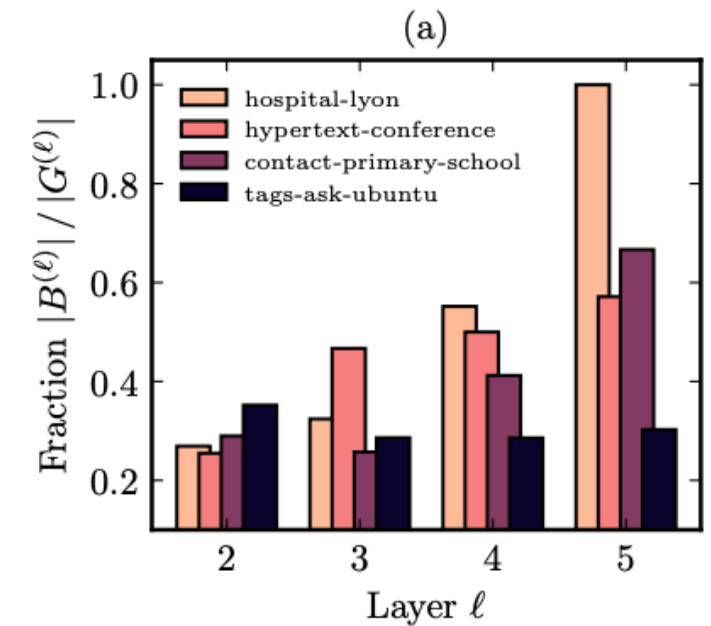
(weighted objective; larger γ values impose stringent weight threshold for keeping edges)

$$\mathcal{L}_w(G, B) = \mathcal{L}(G, B) + \sum_{e \in G} \mathcal{L}(w(e), b_e)$$

Real-world hypergraph backbones

Backbone statistics (left) and edge retrieval (right).

Dataset	N	$ G $	$ B $	η
coauth-mag-geology-1980	1674	903	310	0.80
coauth-mag-geology-1981	1075	547	192	0.82
coauth-mag-geology-1982	1878	987	345	0.80
coauth-mag-geology-1983	1734	883	313	0.81
contact-high-school	327	7818	2223	0.65
contact-primary-school	242	12704	3579	0.70
dawn	2290	138742	52704	0.83
email-enron	143	1459	309	0.61
email-eu	986	24520	5484	0.60
hospital-lyon	75	1824	545	0.69
hypertext-conference	113	2434	691	0.74
invs13	92	787	222	0.77
invs15	217	4909	1415	0.70
kaggle-whats-cooking	6714	39224	8918	0.78
malawi-village	84	431	125	0.68
ndc-classes	628	796	226	0.54
ndc-substances	3414	6471	1549	0.60
science-gallery	410	3350	924	0.64
sfhh-conference	403	10541	2989	0.70
tags-ask-ubuntu	3021	145053	43770	0.73
tags-math-sx	1627	169259	57071	0.75



Summary

Principled, efficient, and interpretable framework for dimensionality reduction of higher-order networks.

Future directions

Extensions of structural reducibility to weighted, directed hypergraphs.

Backboning temporal hypergraphs in downstream applications to contagion processes.

Thanks!

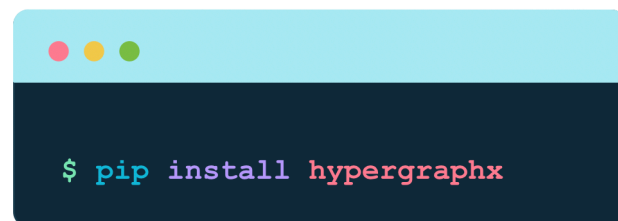


A. Kirkley, H. Felipe, F. Battiston, Structural reducibility of hypergraphs.
Physical Review Letters **135**, 247401 (2025). [Editor's suggestion]



A. Kirkley, H. Felipe, F. Malizia, and F. Battiston, Hypergraph backboning.
[arXiv:2606.00893](https://arxiv.org/abs/2606.00893) (2026).

Data and code available as part of the library Hypergraphx.



Code



Data

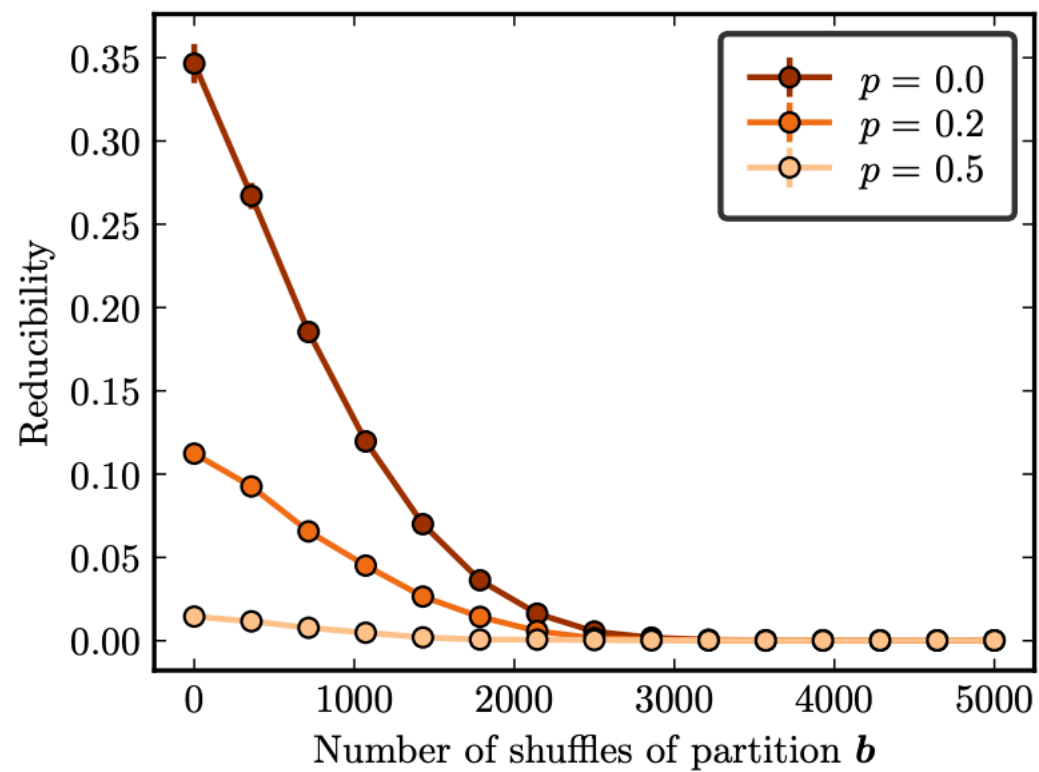
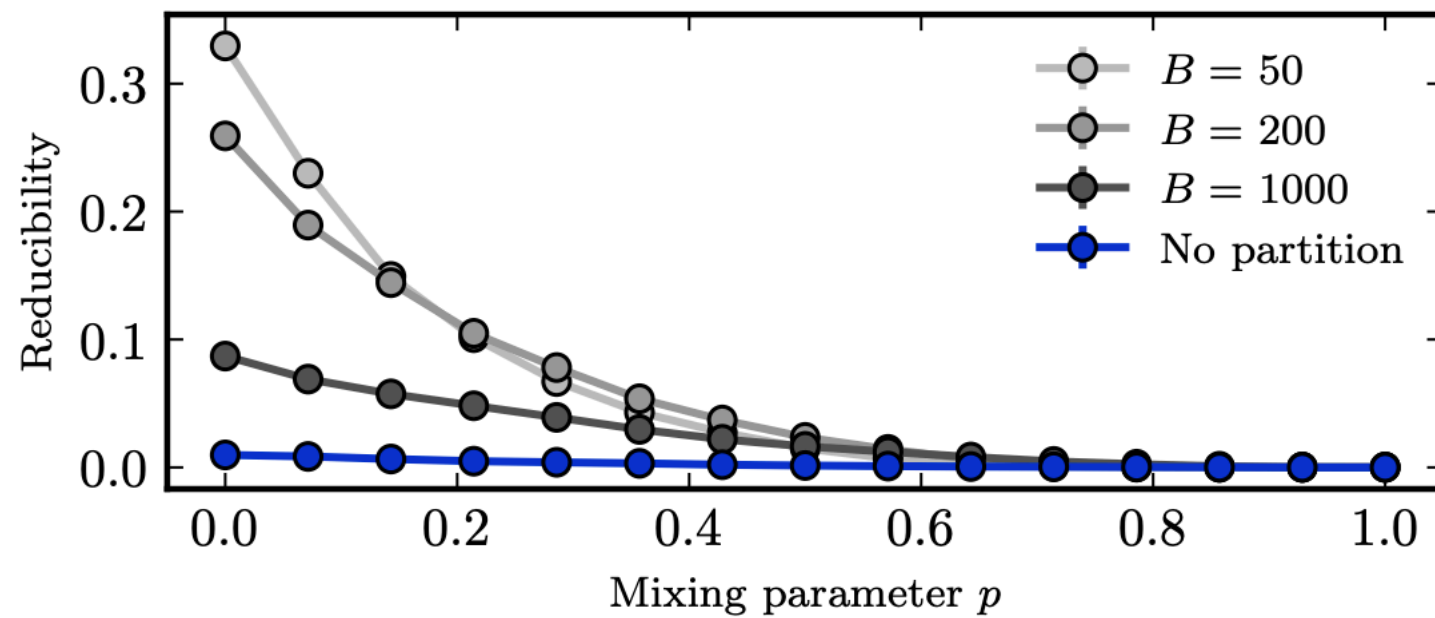


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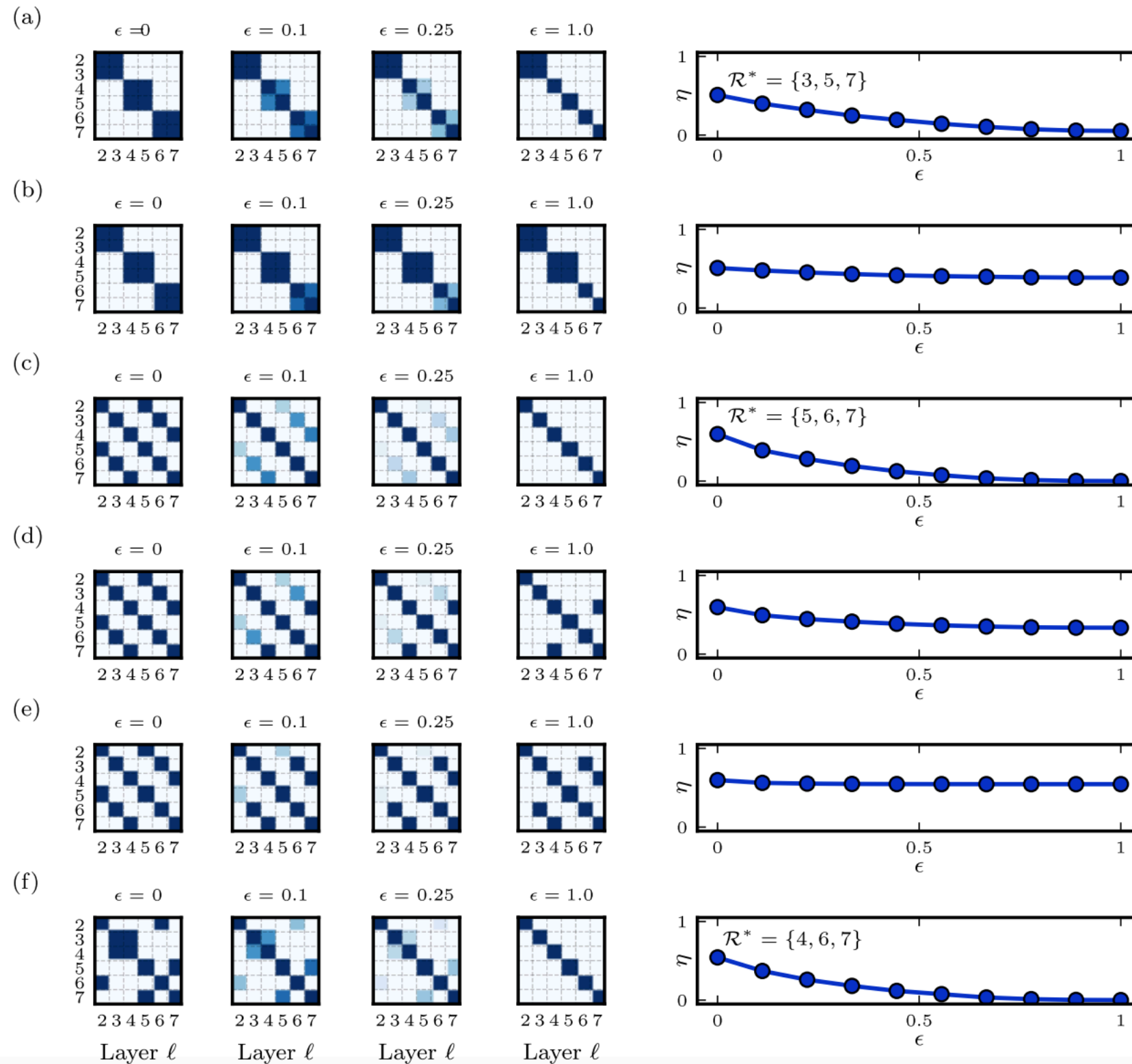


Federico Battiston
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Back-up slides



Back-up slides



Back-up slides

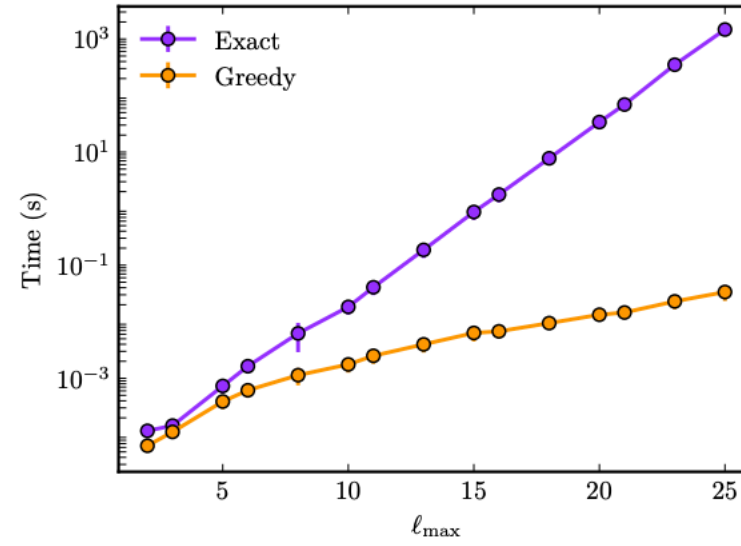
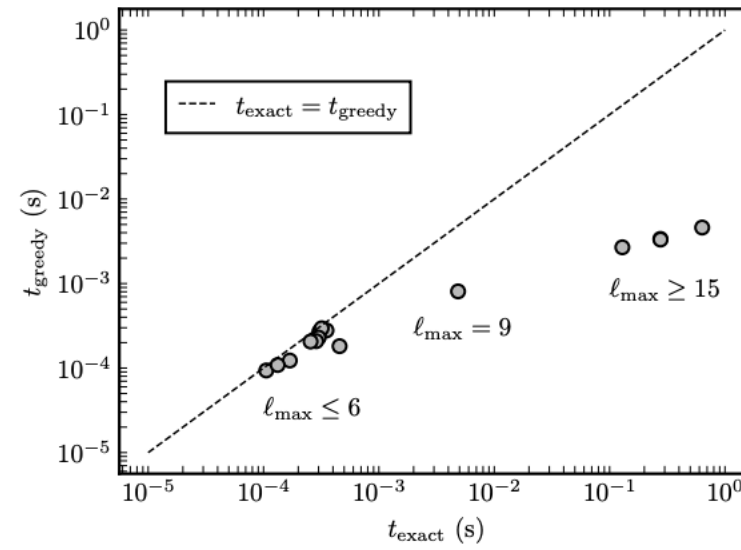


FIG. S5. Runtimes for computing $\mathcal{R}^*$ and η in synthetic fully nested hypergraphs, using both the exact and greedy optimization schemes.



Back-up slides

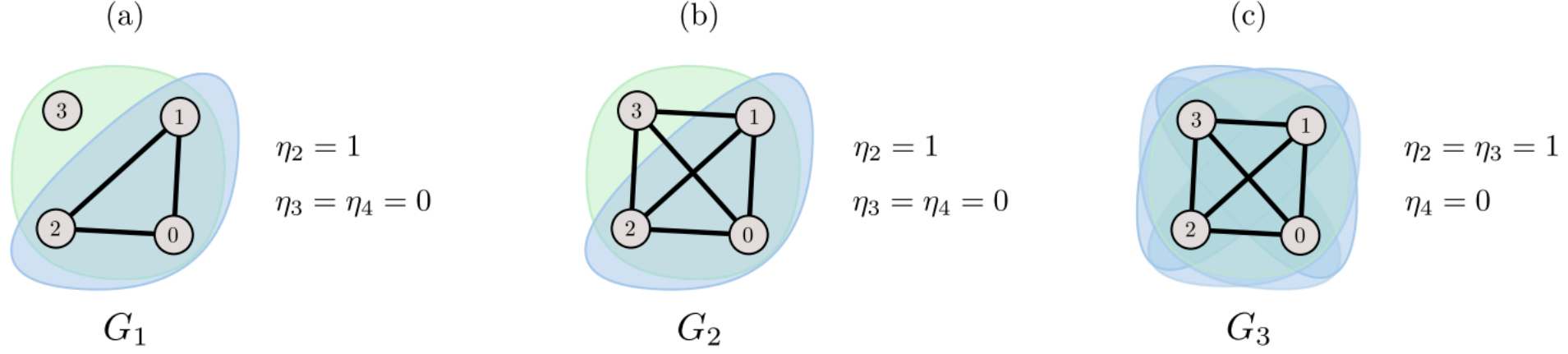


FIG. S7. Small example networks to illustrate layer-wise reducibility. Edges are represented as thick lines, 3-body interactions are blue rounded triangles, and 4-body interactions are green rounded squares. (a) Hypergraph G_1 , with layer $\ell = 2$ fully redundant given $\ell = 3$, but $\ell = 3$ not redundant. (b) Hypergraph G_2 , with layer $\ell = 2$ fully redundant given $\ell = 4$, but $\ell = 3$ again not redundant. (c) Fully nested hypergraph G_3 , with layer $\ell = 2$ fully redundant given $\ell = 3, 4$ and layer $\ell = 3$ fully redundant given $\ell = 4$.

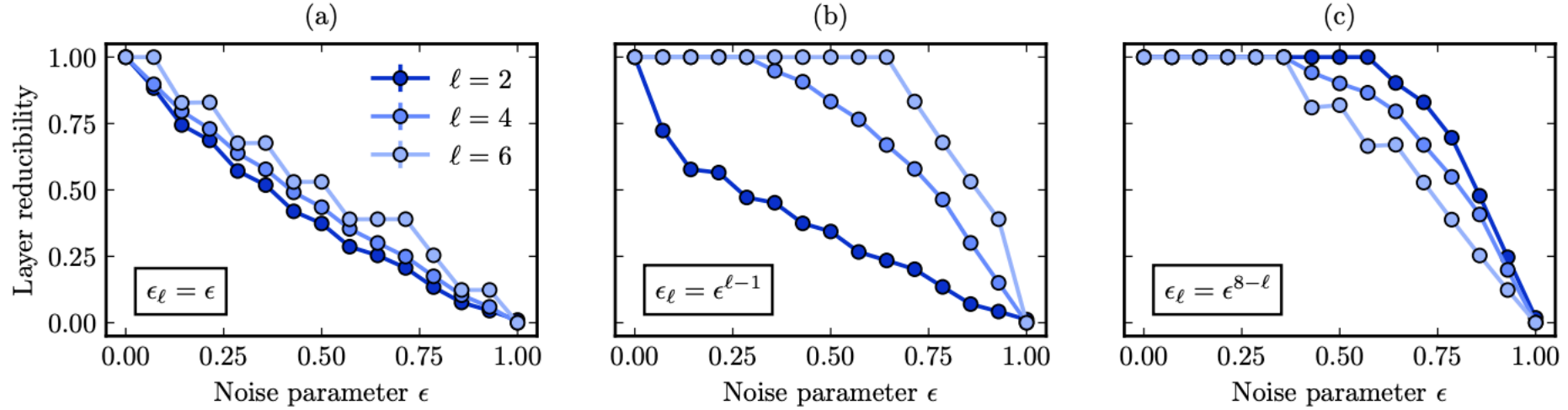


FIG. S8. Layer-wise reducibility of noisy synthetic hypergraphs. Individual layer reducibility values for $\ell = 2, 4, 6$ versus noise level ϵ for: (a) Fully nested hypergraphs with constant noise schedule $\epsilon_\ell = \epsilon$; (b) Fully nested hypergraphs with noise schedule $\epsilon_\ell = \epsilon^{\ell-1}$ more heavily perturbing lower order layers; (c) Fully nested hypergraphs with noise schedule $\epsilon_\ell = \epsilon^{8-\ell}$ more heavily perturbing higher order layers. Experiments are repeated over ten trials and error bars represent two standard errors in the mean.

Back-up slides

